Sahaj Bhadja

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Summary

Software engineer with experience developing **embedded and safety-oriented software** in **C, C++, Rust, and Python**. Background in **simulation, modeling, fault tolerance**, and verification including **unit/integration testing** and **CI/CD**. Passionate about aircraft technology and building reliable flight software systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Embedded Systems, Operating Systems, Distributed Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Embedded Systems, Operating Systems, Software Engineering, Robotics Fundamentals

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed reliable, maintainable code in **C++ and Python** for safety-oriented systems, contributing to **unit tests, integration tests**, and **CI/CD pipelines** with active code reviews.
- Supported **simulation and modeling** workflows and documented designs, requirements, and test results for cross-functional engineering teams.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Implemented embedded-style control modules and backend services with strong focus on **fault mitigation**, logging, and automated test development in collaborative team environment.
- Built hardware/software test infrastructure for validation activities, improving traceability and report generation for release readiness.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Contributed to **cyber-physical simulation** pipelines modeling vehicle dynamics and Monte Carlo scenario analysis for performance and reliability studies.
- Participated in verification activities and documentation of test outcomes, communicating technical results clearly in English to multidisciplinary stakeholders.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Wrote production-quality **C++ and Python** for distributed real-time systems with emphasis on reliability, monitoring, and **fault tolerance** mechanisms.
- Collaborated cross-functionally on requirements, architecture, and system-level testing while maintaining clear design and operational documentation.

AeroFlight Systems **India**
Embedded Software Intern *January 2023 – April 2023*

- Developed **embedded software** for microcontroller and **RTOS** environments in **C/C++**, implementing aircraft-style control functions and critical monitoring logic.
- Supported simulation tools and automated test systems, learning safety-critical development practices including structured V&V and certification-oriented documentation.

Skills

Languages: C, C++, Rust basics, Python, Bash
Embedded: Microcontrollers, bare-metal concepts, RTOS exposure, embedded OS fundamentals
V&V: Unit testing, integration testing, CI/CD, automated testing, simulation support
Systems: Fault tolerance, modeling, documentation, Git, cross-functional collaboration

Projects

FlightSim Control | *C++, Python, Simulation, RTOS*

- Built **6-DoF aircraft simulation** and control prototype with fault injection tests, supporting modeling of flight dynamics and validation of control logic.

FaultGuard Embedded | *C, Rust, Unit/Integration Testing*

- Implemented **fault tolerance and mitigation** modules for embedded targets with comprehensive unit tests, CI pipelines, and structured design documentation.

DriveEval | *Python, C++, Simulation, CI/CD*

- Developed simulation test framework for scenario-based evaluation, data collection, and automated report generation for safety-critical workflow analysis.